Project 3: **Developing a DB from scratch**

CS 347 - Spring 2026

The goals of this project are to design a new database, to implement the database on a MySQL instance, and then to write and execute SQL statements to both populate and use the database for operational reports. Each step below will indicate what you are asked to submit as deliverables of the project (all via the Canvas assignment).

1. Designing a database for a Public Works Department

Consider that you have been hired by the Austin Department of Public Works. You are part of the team who is developing the Pothole Tracking and Repair System (PTRS). The PTRS will be a web-based system where all data entry and queries will be made interactively. Below is the system's description. Your area of the project is the system's database.

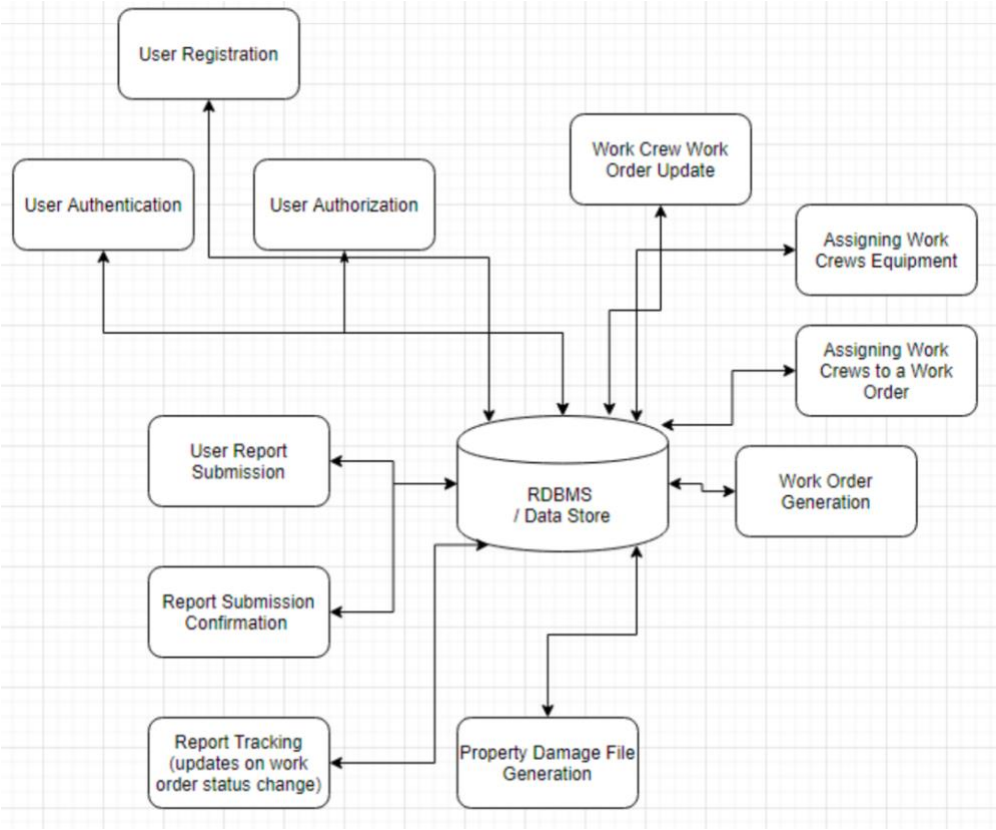
Users will log onto the website and report the location and severity of potholes. As potholes are reported, they are logged within a "PTRS database" (the DB). Potholes are assigned an identifying number by the system, and have the following data: street address, pothole size (on a scale of 1 to 10), location on the street (middle, curb, etc.), district (determined from street address), and repair priority (automatically calculated from the size of the pothole). [*]

*: If your DB design work, later on, requires more detailed information for the above description, it is OK to propose how the system will determine: (a) the district from the street address, maybe by using a table or a function in the app, and (b) the repair priority from the pothole size, or alternatively using the pothole size together with the street address or the district. Notice that the specific choices for these features are left up to you; we will accept your proposal as long as your design is reasonable, and your choices are justified in your report.

A work order will be associated with each pothole, and will include pothole location and size, repair work crew identifying number, number of people on crew, equipment assigned, hours applied to repair, pothole status (in progress, repaired, not repaired), amount of filler material used, and cost of repair (computed from hours applied, number of people, material and equipment used).

Finally, a property damage "file" (a record really, part of the DB) will be created to hold information about any reported damage due to the pothole, and includes citizen's name, address, phone number, type of damage, and dollar amount of damage.

The following *system architectural design* diagram has been created by the development team. As you can see, at the center of the system is the relational database and there are eleven software components identified for the system functionality. Note that this is not an E-R Diagram.



Your first task is to follow the Relational Database model to design the database so that it will support this system. Define what data each entity should hold and how the different entities should have their relationships set up. Specifically, these are the necessary items for this step:

- Define the tables that will make your database up, with their column names and data types. Make sure the table and column names represent the entities and attributes appropriately. Be sensible with regards to the *number* of attributes you define for each entity. Define the primary keys as needed. **Take care to ensure all tables are in BCNF.**
- Define the obvious relationships between the tables and set up the respective foreign keys as needed.
- Develop an Entity-Relationship (E-R) diagram to document the database design.
    - Use Lucidchart to create the E-R diagram with the table information (the design of each table) you defined above.
    - Make sure you include all column names, keys, and data types, in each table. Draw the proper relationships between the tables. Add both minimum and maximum cardinality on the edges of the E-R diagram.
    - Name the diagram appropriately and download it to your computer.

Submit a pdf report file (**report3.pdf**) to the Canvas assignment. In it, describe your database design and the choices you made, indicating how the tables will be used by the application software, and include your E-R diagram in it.

2. Implementing the PTRS database on your MySQL instance

- Produce the DDL statements (in a SQL file) to create the DB and all of its database objects per the system design shown above (name the DB as you wish but be sensible). Pay attention to column data types, integrity constraints, etc.
- Run your DDL file in MySQL to get the DB ready for data population.

Submit this SQL file, to the Canvas assignment.

3. Populating the database on your MySQL instance

- Enter test data of your own for all tables (in sufficient amounts, see *Note below) into appropriate **CSV files**. Be sensible about the minimum number of rows you load into each table; keep in mind that the more data, or the more interesting data, you have in the tables, the better your queries will be tested.
    - *Note: A useful criteria for deciding the amount of test data is: Based on the queries you define for your reports, we expect them to display at least 3 rows of results. The amount of data you need depends on what data the queries are looking for. One row of results would be OK for queries using aggregates.
- Populate the tables using these CSV files. The necessary command is described in the MySQL documentation.

Submit the CSV files, to the Canvas assignment.

4. Making the database operational

a) Develop a set of **six** or more SQL queries to support the required operational reports on your DB, based on the tables you defined and created, and the structure shown in the system diagram above. Notice that the diagram uses the word "Report" to refer to the submissions from the external users (the concerned citizens) and not to the operational reports needed by the managing organization.

In the system diagram above, you should notice that **SQL query reports** will be needed for: users, pothole reports, work orders, property damage, work crews, and work equipment.

b) Create the SQL statements for the above query reports and store them in a SQL file (**project3.sql**). Make sure this file has comments that document which report is implemented by each statement.

Submit this SQL file, to the Canvas assignment.

c) Execute your queries to show they are working correctly with the data stored in the DB.

Submit screen shots **of the results** of executing each of the above queries, to the Canvas assignment. They should be combined into a pdf file, **results3.pdf**.

5. See the grading rubric below.

Project 3 Rubric

Create the database design file, **report3.pdf**, including the E-R diagram. **-15** no E-R diagram, **-10** incomplete diagram **-2** incomplete definition for each table (purpose, etc.)	25
Create the DDL file and run it on the MySQL instance. **-10** errors in the DDL code **-4** incomplete implementation for each DB table (data types, constraints, etc.)	15
Populate the DB tables from CSV files. See note in section #3 above. **-4** for each table missing data **-2** for each table with insufficient data, or missing CSV file	20
Create the DML file for implementing **six** or more SQL queries. Code is to be found in the **project3.sql** file. Include a short comment above each SQL statement to explain the purpose of each report. **-3** for each missing query or if a report is not operationally useful **-1** for each missing comment or if comment is not descriptive	20
Run the **six** or more queries for the above reports. Include the screen shots in a results file. Output of these queries is to be found in the **results3.pdf** file. **-3** for each missing query execution screen shot	20
Total Credit:	**100**